NOT MEASUREMENT SENSITIVE

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DETAIL SPECIFICATION

SEALING COMPOUND, GASKET, HYDROCARBON FLUID AND WATER RESISTANT

REACTIVATED AFTER 10 JULY 2026 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers requirements for thread and gasket sealing compounds that are resistant to fuel, engine oils, and water.

1.2 Classification. This specification covers three types of sealing compounds:

Type I - Hardening
Type II - Nonhardening
Type III - Nonhardening (low viscosity)

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

COMMERCIAL ITEM DESCRIPTIONS

A-A-52624	-	ANTIFREEZE, MULTI-ENGINE TYPE

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product

FEDERAL SPECIFICATIONS

PPP-C-96	-	Can, Metal, 28 Gage and Lighter

FEDERAL STANDARDS

FED-STD-123	-	Marking for Shipment (Civil Agencies)
FED-STD-313	-	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities.

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF TRANSPORTATION (DOT)

49 CFR 100-199	-	Rules and Regulations for the Transportation of Hazardous Materials

(Copies of this document are available online at https://www.ecfr.gov/current/title-49.)

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OSHA REGULATIONS - HAZARDOUS COMMUNICATION STANDARDS

29 CFR 1910.1200	-	Hazard Communication

(Copies of this document are available from https://www.osha.gov.)

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM A48/A48M	-	Standard Specification for Gray Iron Castings
ASTM A109/A109M	-	Standard Specification for Steel, Strip, Carbon, (0.25 Maximum Percent), Cold-Rolled
ASTM B121/B121M	-	Standard Specification for Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM B152/B152M	-	Standard Specification for Copper Sheet, Strip, Plate, and Rolled Bar
ASTM B210/B210M	-	Standard Specification for Aluminum and Aluminum-Alloy Drawn Seamless Tubes
ASTM B545	-	Standard Specification for Electrodeposited Coatings of Tin
ASTM D56	-	Standard Test Methods for Flash Point by Tag Closed Cup Tester
ASTM D471	-	Standard Test Method for Rubber Property–Effect of Liquids
ASTM D638	-	Standard Test Method for Tensile Properties of Plastics
ASTM D1084	-	Standard Test Methods for Viscosity of Adhesives
ASTM D2556	-	Standard Test Method for Apparent Viscosity of Adhesives Having Shear-Rate-Dependent Flow Properties Using Rotational Viscometry

(Copies of these documents are available online at https://www.astm.org/.)

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SAE INTERNATIONAL

AMSQQA250/4 - Aluminum Alloy 2024, Plate and Sheet

AS5206 - Nut, Flared, Long

(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Solubility. The solubility of the sealing compound shall be as specified in table I when tested in accordance with 4.6.2. The aromatic fluid and ethylene glycol shall conform to the specifications in table I. The solubility of type III material in water shall be tested in accordance with 4.6.8.

TABLE I. Solubility.

Fluid	Type I % max	Type II % max	Type III % max
Aromatic fluid, ASTM D471, Ref Fuel D	12	10	15
Aromatic fluid, ASTM D471, Ref Fuel B	8	9	12
Ethylene glycol, A-A-52624	8	5	13
Distilled water	4	6	2
ASTM D471 IRM 901 Oil	8	10	4
ASTM D471 IRM 902 Oil	8	10	4
ASTM D471 IRM 903 Oil	8	10	4

3.2 Torque release. The torque release shall not be more than the following applicable maximum percent increase when tested in accordance with 4.6.3:

Type I - 100 percent

Type II - 70 percent

Type III - 40 percent

3.3 Volatile content. The volatile content shall be as follows when tested in accordance with 4.6.4:

Type I - 15.0 ± 3 percent

Type II - 13.0 ± 3 percent

Type III - 24.0 ± 3 percent

3.4 Flow test (vertical). Flow shall not be more than 0.25 inch (6.3 mm) when tested in accordance with 4.6.5.

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3.5 Viscosity. The viscosity of the sealing compound shall be as specified in table II when tested in accordance with 4.6.6.

TABLE II. Viscosity.

Type I % max	Type II % max	Type III % max
4,000 - 20,000 poises	4,000 - 20,000 poises	30 - 680 poises (3,000 - 68,000 centipoises)

3.6 Ash content. The ash content of the sealing compound shall be as follows when tested in accordance with 4.6.7:

Type I - 48 ± 5 percent
Type II - 48 ± 5 percent
Type III - 15 ± 5 percent

3.7 Water-soluble material (type III only). The nonvolatile portion of the compound shall not be more than 2 percent soluble in water when tested in accordance with 4.6.8.

3.8 Corrosion. The following metals (of the specification as shown in table III) after 5 day treatment (immersion) with the compound as specified in 4.6.9, shall not exceed the maximum change in weight (loss or gain) as listed in table III.

TABLE III. Corrosion loss or gain in weight.

Test strip specification	Maximum loss or gain in milligrams per square centimeter (mg/cm^2)		
	Type I	Type II	Type III
Tinned Iron	0.7	0.7	0.7
Steel	0.7	0.7	0.7
Aluminum Alloy	0.1	0.1	0.1
Cast Iron	1.2	1.0	1.2
Brass	0.3	0.3	0.3
Copper	0.2	0.2	0.2

3.9 Flash point. The flash point of the sealing compound shall not be less than 10°C (50°F) when tested in accordance with 4.6.10.

3.10 Flexibility (types II and III only). The flexibility of the sealing compound shall be as follows when tested in accordance with 4.6.11:

Type II sealing compound shall not crack when flexed at -34 ± 1°C (-30 ± 2°F).
Type III sealing compound shall not crack when flexed at -34 ± 1°C (-30 ± 2°F).

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3.11 Consistency. Types I and II sealing compound shall be of such consistency that it may be applied with a spatula or stiff brush (see 4.5.1). Type III sealing compound shall be of such consistency that it may be applied with an ordinary paint brush.

3.12 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief:

(http://www.osha.gov/dsg/hazcom/osha-brief.html).

and, as specified in Appendix D of 29 CFR 1910.1200 see:

(https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.1200AppD).

A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.6). The SDS shall be included with each shipment of the material covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

3.12.1 Toxicity. The gasket sealing compound shall have no adverse effect on the health of personnel (see 6.3).

3.13 Workmanship. The compound shall be a smooth and homogenous mixture, free from lumps, caked material, and particles of foreign matter.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. All inspection shall be to determine compliance with the requirements of this specification to serve as a basis for acceptance.

4.3 Lot. A lot shall consist of sealing compound from the same manufacturer's batch. A batch is defined as that quantity of material which has been subject to some unit chemical or physical mixing process intended to make the final product substantially uniform.

4.4 Sampling.

4.4.1 Examination. Unless otherwise specified, a random sample of filled containers shall be selected for examination in accordance with MIL-STD-1916 and surveyed under level S-1 of ANSI/ASQ Z1.4.

4.4.2 Tests. A sample consisting of one gallon of sealing compound shall be taken from each lot for tests.

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4.5 Examination.

4.5.1 Visual. Sample units selected in accordance with 4.4.1 shall be examined for the defects and at the acceptable quality level shown is table IV.

TABLE IV. Methods of examination.

Material	AQL Percent	Classification of defect	Defect	Method of inspection
Sealing compound (see 3.13) (see 3.11)	2.5	Major 101 Major 102 Major 103 Major 104	Not homogeneous Lumps or caked material Foreign matter Wrong consistency [1/]	Visual Visual Visual Visual
Container, open (see 5.1 and 6.5.1.1)	2.5	Major 105 Major 106	Improper type Improper fill [2/]	Visual Approved scale [3/]
Container, closed (see 5.1 and 6.5.1.1)	2.5	Major 107 Major 108	Leakage Improper closure	Visual Visual
Container, closed (see 5.1 and 6.5.1.1.1.3)	2.5	Major 109	Missing brushes (if applicable)	Visual

[1/] The sealing compound and surroundings shall be at room temperature (70° to 90°F).
[2/] The actual weight of a container filled with the minimum required quantity of sealing compound shall be the basis for determining the acceptable weight of subsequent containers.
[3/] Approved by procuring activity.

4.5.2 Packaging, packing and marking for shipment. Examination shall be made for the defects at the acceptable quality level (AQL) shown in table IV to determine compliance with Section 5 and possible legacy requirements (if cited in the contract or purchase order) under 6.5.

4.6 Tests.

4.6.1 Test specimens. Unless otherwise specified, the sealing compound shall be tested without treatment or preconditioning of any kind and tests shall be conducted at a temperature of 240 ± 2°C (75°C ± 5°F) and a relative humidity of 50 ± 5 percent.

4.6.2 Solubility. Each of the fluids in table V shall be used separately as the immersion medium at the indicated temperatures.

TABLE V. Immersion media and temperatures.

Fluid	Type I % max	Type II % max	Type III % max
Aromatic fluid, ASTM D471, Ref Fuel D	12	10	15
Aromatic fluid, ASTM D471, Ref Fuel B	8	9	12
Ethylene glycol, A-A-52624	8	5	13
Distilled water	4	6	2
ASTM D471 IRM 901 Oil	8	10	4
ASTM D471 IRM 902 Oil	8	10	4
ASTM D471 IRM 903 Oil	8	10	4

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4.6.2.1 Test panels. Three recessed aluminum panels, conforming to the design and measurement of figure 1, shall be used to test the solubility of each type of sealing compound in each immersion medium. The test panels shall be degreased with a suitable degreasing solvent. Care shall be exercised to avoid skin contact, inhalation, and ingestion of vapors. The panels shall then be placed in an oven at 70 ± 1°C (158 ± 2°F) for 1 hour, cooled in a desiccator for 1 hour, and then weighed to the nearest 1 mg.

4.6.2.2 Procedure. The test panels (see 4.6.2.1) shall be filled to the rim of recess with the sealing compound and dried in a circulating air oven for 48 hours at 70 ± 2°C (158 ± 2°F). After removal from the oven and cooling to room temperature in a desiccator, the specimens shall be weighed to the nearest 1 mg and then immersed for 24 hours in 250 ml of the respective fluids. For the aromatic fluids the panels shall be placed horizontally in the containers to prevent flow of the compounds. For the boiling water container, a reflux condenser is necessary to prevent loss of water. All immersion containers shall be covered to prevent loss of immersion fluid by evaporation. After ethylene glycol immersion, excess glycol shall be removed by immersing the specimen in water at 23 ± 2°C (74 ± 5°F) for 15 min. After immersion, all specimens shall be dried in a circulating air oven at 70 ± 1°C (158 ± 2°F) for 24 hours. After removal from the oven, the specimens shall be cooled to room temperature in a desiccator for 1 hour before weighing. Test panels shall be reweighed and percent change in weight shall be calculated as follows:

$$\text{Percent change in weight} = \frac{W_1 - W_2}{W_1 - W_P} \times 100$$

Where:

W_1 = weight of the panel plus sealing compound before immersion.

W_2 = weight of the dried panel plus sealing compound after immersion.

W_P = weight of panel.

4.6.2.3 Report. Results shall be reported as the average of 3 calculations of percent change in weight for each type of sealing compound and immersion medium.

4.6.3 Torque release (test fittings). Three hydraulic fitting assemblies comprised of a union, a corresponding SAE-AS5206 specified nut, and a sufficient length of 0.5 in flared aluminum tubing conforming to ASTM B210/B210M shall be utilized for this test. The sealing compound shall be brushed on the threads of the clean union to fill the threads approximately level, and then assembled with nut and tubing. After assembly has been completed, excess compound which has extruded shall be wiped from the flange. The assembly shall be tightened to 200 in-lbs torque and placed in a circulating air oven for 7 days at 700 ± 1°C (158 ± 2°F). After removal from the oven, excess compound on the outer edge of the joint shall be removed by scraping and the assembly shall be placed in a desiccator at room temperature for 24 hours. Torque required to break the seal shall then be determined with the same wrench used for the assembling of the fittings. Results shall be reported as the average of the three readings.

4.6.4 Volatile content. Approximately 5 g of the compound shall be placed in an accurately weighed flat-bottom, metal dish having a diameter of about 6 cm and depth of about 2 cm. The dish with the compound shall be weighed and then placed in a circulating air oven at 70° ± 1°C (158 ± 2°F) for 48 hours. After removal from the oven, the dish shall be cooled to room

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temperature in a desiccator for 1 hour and then reweighed. Percent of volatile content shall be calculated as follows:

$$\text{Volatile content, percent} = \frac{W_1 - W_2}{W_1 - W_D} \times 100$$

Where:

W_1 = weight of dish and sample before drying.

W_2 = weight of dish and sample after drying.

W_D = weight of empty dish.

4.6.5 Flow test (vertical).

4.6.5.1 Types I and II. Three aluminum alloy panels conforming to SAE-AMS-QQ-A-250/4 approximately 1 in by 6 in (25 mm by 150 mm) shall be coated over a 1 in by 2 in (25 mm by 50 mm) section with 0.125 in coating of sealing compound. A scribe line shall be made on the uncoated surface of the metal 0.25 in (6mm) below the compound. Immediately after coating, the panels shall be placed in the vertical positions with the scribe line at the bottom and allowed to stand in this position for 1 hour at 70° ± 10C (158 ± 2°F) to determine the flow of the compound in reference to the scribe line. Results shall be reported as the average flow of the three readings.

4.6.6 Viscosity. The viscosity of sealing compound shall be determined in accordance with ASTM D 2556 for non-Newtonian materials (i.e. shear-rate-dependent materials). The viscosity of sealing compound shall be determined in accordance with ASTM D 1084 (Procedure B) for materials that have Newtonian flow characteristics (i.e. viscosity independent of shear stress or shear rate).

4.6.7 Ash content. Weigh about 2 g of the sealing compound in a previously weighed porcelain crucible. Heat the crucible and contents to approximately 925°C (1700°F) in an electric muffle furnace, or if not available, a gas burner yielding a similar temperature, is also suitable. Cool in a desiccator for 1 hour and reweigh. Reheat to approximately 925°C (1700°F) and cool in a desiccator until a constant weight is obtained. Calculate the percent of ash as follows:

$$\text{Percent Ash} = \frac{A - B}{C - B} \times 100$$

Where:

A = weight of residue and crucible.

B = weight of crucible.

C = weight of sample and crucible.

4.6.8 Water-soluble material (type III only). The determination of water-soluble material in type III sealing compound shall be as follows:

Place 10 g of sealing compound in a 400 ml beaker and add 100 ml of distilled water. Boil the mixture for 10 min and filter while hot through No. 40 Whatman or equivalent ashless filter paper into a previously weighed porcelain evaporation dish (W_o). Evaporate the filtrate to dryness at a

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temperature just below the boiling point of water (100°C) (212°F). Heat the dish and contents in an air circulating oven at 100 ± 2°C (212 ± 3°F) for 1hour, cool in a desiccator to room temperature, weigh (W_f). Calculate the percent of water-soluble material as follows:

EQ 1 Percent water-soluble material = $\frac{W_n}{W_s - W_v} \times 100$

Where:

W_n = weight of nonvolatile component which is water-soluble.

W_s = weight of sealing compound (original sample).

W_v = weight of volatile component (see 4.6.4).

EQ 2 $W_n = W_f - W_o$

Where:

W_o = weight of dish (original).

W_f = weight of dish containing residue after heating.

EQ 3 $W_v = XW_s$

Where:

X = % volatile content.

4.6.9 Corrosion. Test strips for the corrosion test shall conform to the following specifications as specified by the contracting officer.

a.	Tinned Iron	ASTM B545
b.	Steel	ASTM A109/A109M
c.	Aluminum Alloy	AMSQQA250/4
d.	Cast Iron	ASTM A48/A48M
e.	Brass	ASTM B121/B121M
f.	Copper	ASTM B152/B152M

The test strips shall have dimensions of 8.8 cm x 1.2 cm x 0.3 cm and a surface area of approximately 22 cm^2. The strips (with the exception of tinned iron) shall be polished to remove pit burrs and irregularities from the faces and edges and finished with a 240 grit polishing medium. The tinned iron shall be free from pit burrs and irregularities and shall be polished with a clean cloth. All specimens shall be held in a suitable holder to avoid contact with the hands. After polishing, the specimens shall be cleaned by swabbing in hot naptha and a final rinse in warm methanol. Follow this by drying in an air circulated oven at 66°C (150°F) for 2 hours. After drying, weigh the strips to the nearest 0.1 mg. The weighed strips shall be arranged in the order of metals as above. The strips shall be joined in a metal-to-metal contact by fastening with a steel bolt that passes through holes drilled near one end of each strip. The strips are so bent, that except for one small terminal area near the bolt, the strips are not in contact one with another. The strips

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shall then be immersed into the sealing compound for a period of 120 hours (5 days) at 66°C (150°). At the end of the test period, remove the strips and wash thoroughly with isopropyl alcohol and follow with an acetone rinse. A soft hairbrush may be used in this process. The cleaned strips shall be oven dried thoroughly at 66°C (150°), cooled to room temperature in a desiccator, and weighed to the nearest 0.1 mg. Any changes in weight shall be attributed to corrosion.

4.6.10 Flash point. The flash point of the sealing compound shall be determined in accordance with ASTM D 56.

4.6.11 Flexibility (types II and III only). Prepare two smooth brass panels conforming to ASTM-B121/B121M, 6 in x 2.5 in x 0.025 in (152 mm x 53 mm x 0.6 mm) and clean with acetone. Apply a 0.002 in (0.05 mm) thick coat of sealing compound across one surface of each panel, using a doctor blade. Allow to air dry 6 days at room temperature, then 2 hours at 70 ± 1°C (158 ± 2°F). Prepare two panels for each material being tested. Panels coated with type II sealing compound shall be cooled at -34 ± 1°C (-30 ± 2°F) for 4 hours prior to test. Panels coated with type III sealing compound shall be cooled at -34 ± 1°C (-30 ± 2°F) for 16 hours prior to test. While still at the test temperature and with the coated side outward, rapidly bend the test specimens through an angle of 180 degrees over a mandrel of 0.125 in (3.1 mm) diameter.

4.7 Rejection.

4.7.1 Examination. Any nonconforming sample units shall be rejected and if the number of rejected units is greater than the acceptance number for the sample size, the lot shall be rejected.

4.7.2 Tests. A lot shall be rejected for failure to comply with the test requirements when tested in accordance with 4.6.

4.8 Retest.

4.8.1 Examination. Retests shall be permitted in accordance with MIL-STD-1916.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packaging notes. Due to the substantial passage of time since the last revision of this specification, additional packaging information that may be helpful including past definitions and procedures for unit of product, sampling, packaging inspection, and exterior container markings are included in 6.5 and its sections therein. This legacy content will be deleted in the next revision or amendment to this specification.

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5.2 Marking. Unless otherwise indicated in the contract or purchase order, any special marking required by shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The sealing compound is intended for sealing threads and gaskets where resistance to engine oils, fuel, and water is required.

6.2 Ordering data. Purchasers should select the preferred option permitted herein and include the following information in procurement documents:

a. Title, number and date of this specification.
b. Type of sealing compound required (see 1.2).
c. Quantity desired.
d. Whether a SDS is required with each shipment (see 3.12 and 6.6).
e. Selection of applicable levels of preservation, packing and marking. (see 5.1 and 6.5).
f. Size and type of container (see 5.1 and 6.5).

6.3 Toxicity. Any questions raised regarding toxicity should be referred to the Surgeon General who will act as advisor to the procuring activity.

6.5 Packaging considerations. Due to the substantial length of time since the last revision of this specification, the legacy preservation, packing, and marking information requirements including prior definitions for unit of product with respect to flammable and nonflammable products are included as a courtesy in this section and procuring officials should follow current procedures for packaging in 5.1.

6.5.1 Preservation. Preservation should be level A, C, or commercial as specified (see 6.2).

6.5.1.1 Level A.

6.5.1.1.1 Nonflammable liquid.

6.5.1.1.1.1 Tubes. Unless otherwise specified, type I and II compounds should be commercially prepared units packed in collapsible tubes of approximately 1 oz, 2.25 oz, or 8 oz capacity normally used for unit packing this type of product.

6.5.1.1.1.2 Cans. Alternatively, type I and II compounds may be unit packed in 1 pint, 1 quart, or 1 gallon metal cans conforming to PPP-C-96 of type normally used in unit packing this type of compound.

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6.5.1.1.1.3 Type III. Type III compound should be unit packed quarter pint, 1 pint or 1 gallon metal cans conforming to PPP-C-96 of type normally used to unit pack this type of compound. Quarter pint and 1 pint containers should be fitted with a brush applicator.

6.5.1.1.1.4 Container closure. Closure of specified containers should be in accordance with container specifications and appendices thereto as applicable.

6.5.1.1.2 Flammable liquids. When the flash point of the sealing compound is such that it meets the Department of Transportation definition of "Flammable Liquid", a completed SDS, prepared in accordance with FED-STD-313 should be prepared and the compound should be unit packed with exterior containers marked in accordance with the applicable requirements contained herein and any markings required by MIL-STD-129.

6.5.1.2 Level C.

6.5.1.2.1 Nonflammable liquids. The sealing compound should be preserved to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity for immediate use or for controlled humidity storage.

6.5.1.2.2 Flammable liquids. See 6.5.1.1.2.

6.5.1.3 Commercial. When specified, commercial preservation should be in accordance with FED-STD-123 for nonhazardous material.

6.5.2 Packing. Packing should be level A, C, or commercial as specified (see 6.2).

6.5.2.1 Level A.

6.5.2.1.1 Nonflammable liquids. Compound unit packed in commercially prepared collapsible tubes or metal cans as specified in 6.5.1.1.1 should be packed in accordance with commercial practices or PPP-C-96, as applicable.

6.5.2.1.2 Flammable liquids. When the compound is classified as a "Flammable Liquid", a completed SDS, prepared in accordance with FED-STD-313 should be prepared with interior packages and exterior containers marked in accordance with the applicable requirements contained therein and any markings required by MIL-STD-129.

6.5.2.2 Level C.

6.5.2.2.1 Nonflammable liquids. The sealing compound should be packed to afford adequate protection against damage during shipment within the Continental United States direct from the supply source to the first receiving activity. This level as a minimum should conform to applicable carrier rules and regulations.

6.5.2.2.2 Flammable liquids. See 6.5.2.1.2.

6.5.2.3 Commercial. When specified, packing should be in accordance with FED-STD-123 for nonhazardous material.

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6.5.3 Marking.

6.5.3.1 Nonflammable liquids. Interior packages and exterior containers should be marked in accordance with MIL-STD-129. In addition, the following marking should appear on each container:

a. Store in a cool place
b. Date of manufacture
c. Manufacturer's instructions for use (to include thinning directions, if applicable)

6.5.3.2 Flammable liquids. When the compound is classified as a "Flammable Liquid", a completed SDS, prepared in accordance with FED-STD-313 should be prepared with interior packages and exterior containers marked in accordance with the applicable requirements contained therein and any markings required by MIL-STD-129. In addition, the following marking should appear on each container:

a. Store in a cool place
b. Date of manufacture
c. Manufacturer's instructions for use (to include thinning directions, if applicable)

6.5.3.3 Commercial. When specified, marking should be in accordance with FED-STD-123 for nonhazardous material.

6.6 SDS. Contracting officers will identify those activities requiring copies of a completed SDS prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.7 Subject term (key word) listing.

flammable	immersion	viscosity
Fuel resistant	organic	volatile

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - SH

Preparing activity:
Army - MR
(Project 8030-2026-014)

Review activities:

Army - ME
Navy - AS, YD
Air Force - 84
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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